PSU Community Garden Soil Results

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Overview

- 13 plots tested (Figure 1)

Tests:

- Mehlich-3 (M3) soil fertility
- Organic matter content (%)
- Testing done at Ag Analytical

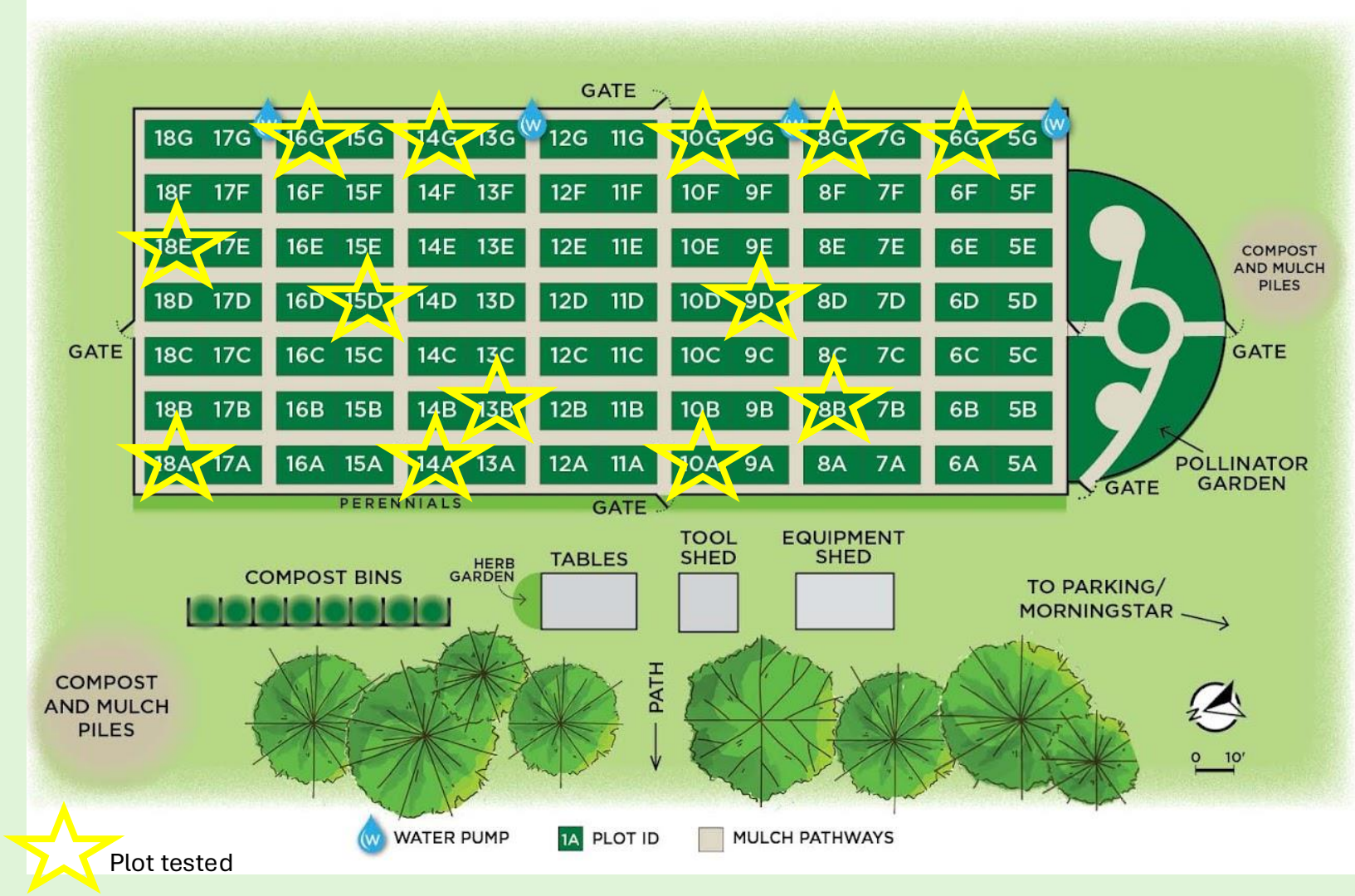

Figure 1. Map identifying plots tested for soil fertility and organic matter.

Mehlich-3 (M3) is a standardized test for nutrient availability

What's included in the test?

- pH
- Phosphorus (P) (ppm)
- Potassium (K) (ppm)
- Magnesium (Mg) (ppm)
- Calcium (ppm)
- Acidity (meq/100g)
- Cation Exchange Capacity (CEC) (meq/100g)
- K% saturation of CEC
- Mg% saturation of CEC
- Ca% saturation of CEC
- Zinc (ppm)
- Copper (ppm)
- Sulfur (ppm)

***If you'd like to read about the methods for each, please refer to:** https://agsci.psu.edu/aasl/soil-testing/methods

Organic matter tests measure the % of decomposed plant and animal material in soil

Why is it important?

- Improves soil structure
- Improves soil drainage
- Supports soil microbes
- Provides nutrients
- Good indicator of productivity

***If you'd like to read about the methods for each, please refer to:** https://agsci.psu.edu/aasl/soil-testing/methods

Results: Tabular Format

	Field ID												
	10G	16G	15D	8G	14G	18E	8B	10A	18A	13B	6G	14A	9D
pH	7.0	7.3	7.1	7.3	7.2	7.2	7.2	7.1	7.2	7.2	6.9	7.2	6.9
Phosphorus (ppm)	343	305	252	270	335	216	304	262	289	272	345	279	338
Potassium (ppm)	290	240	315	177	253	198	240	227	506	375	232	499	361
Magnesium (ppm)	349	385	360	320	342	414	326	357	363	331	392	358	423
Calcium (ppm)	4911	5151	4649	4533	4966	5807	5184	5078	5189	4233	5938	4681	6330
Acidity (meq/100g)	0	0	0	0	0	0	0	0	0	0	0	0	0
CEC (meq/100g)	18.7	18.8	18.8	18.1	18.5	19	18.3	18.6	19.3	18.7	18.9	19.3	19.5
K% Saturation of CEC	4	3.3	4.3	2.5	3.5	2.7	3.4	3.1	6.7	5.1	3.2	6.6	4.8
Mg% Saturation of CEC	15.6	17	16	14.7	15.4	18.2	14.8	16	15.7	14.7	17.3	15.5	18.1
Ca% Saturation of CEC	80.4	79.7	79.8	82.8	81.1	79.1	81.8	80.8	77.6	80.1	79.5	77.9	77.1
Zinc (ppm)	22.8	18.5	17.4	24.8	24.5	20.1	20.7	20.7	22	18.4	28.1	18.1	28.7
Copper (ppm)	2	2	2	1.8	2.1	1.9	1.8	2.5	2.6	2.2	1.8	1.9	2.1
Sulfur (ppm)	15.8	18.2	19.6	14.4	15.9	17.8	17	15.6	25	29.3	19.1	26.1	25.5
Organic Matter (%)	9.9	10.2	9.6	10.1	9.8	12.6	10.5	10.6	11.5	7.4	13.5	9.1	12.6

Results: Statistical Overview

	Average	Std Dev	Min	Max
pH	7.1	0.1	6.9	7.3
Phosphorus (ppm)	293.1	39.8	216.0	345.0
Potassium (ppm)	301.0	106.7	177.0	506.0
Magnesium (ppm)	363.1	32.2	320.0	423.0
Calcium (ppm)	5126.8	593.8	4232.5	6329.9
Acidity (meq/100g)	0.0	0.0	0.0	0.0
CEC (meq/100g)	18.8	0.4	18.1	19.5
K% Saturation of CEC	4.1	1.4	2.5	6.7
Mg% Saturation of CEC	16.1	1.2	14.7	18.2
Ca% Saturation of CEC	79.8	1.6	77.1	82.8
Zinc (ppm)	21.9	3.7	17.4	28.7
Copper (ppm)	2.1	0.3	1.8	2.6
Sulfur (ppm)	19.9	4.8	14.4	29.3
Organic Matter (%)	10.6	1.6	7.4	13.5

pH levels are mostly within optimal ranges for most garden plants

- **Average:** 7.1
- **Min:** 6.9
- **Max:** 7.3
- While some nutrient availability such as iron, copper, and zinc can decrease at pH values >7, (Figure 3), the risk of having problems at the pH levels in the garden are low at 7.3
- Avoid acid-loving crops such as blueberries and azaleas
- Avoid planting potatoes as a pH > 7 can promote a disease called common scab and can reduce yield

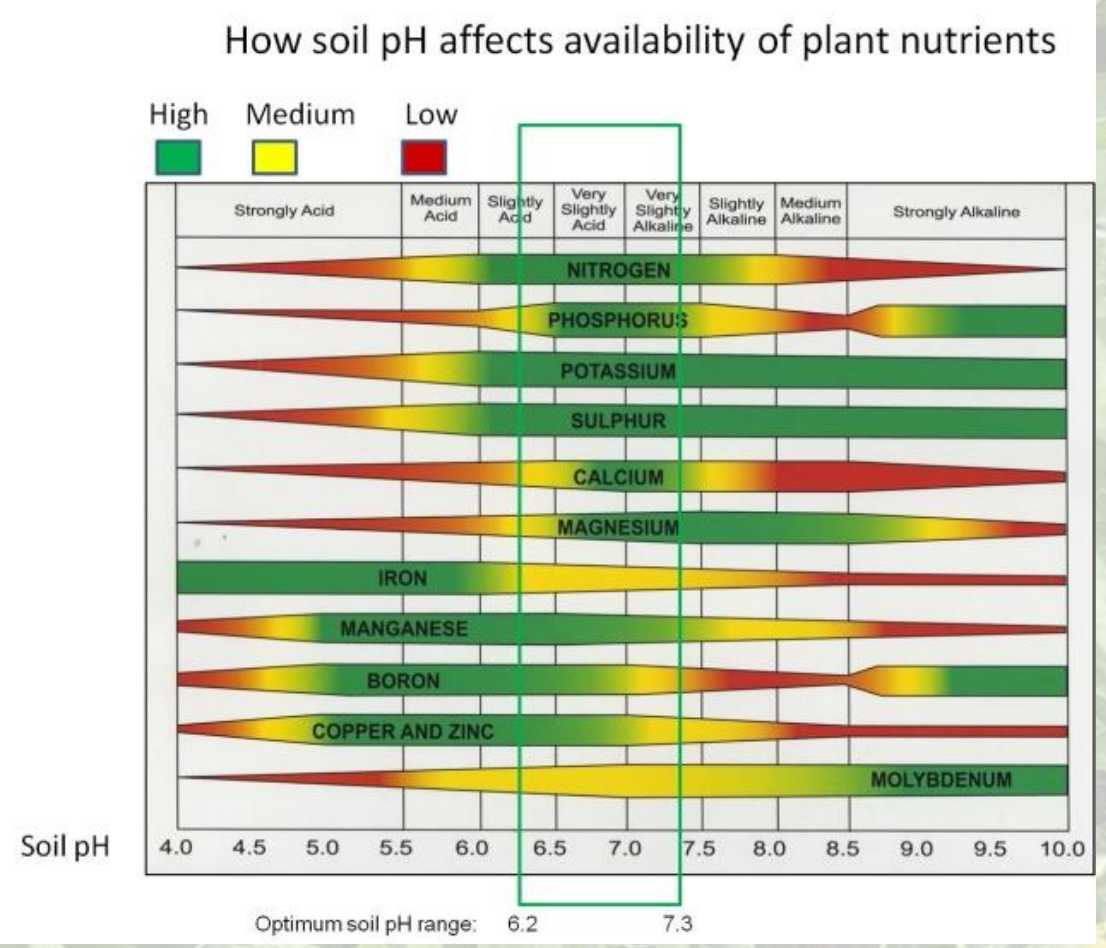

Figure 2. Nutrient availability ranges with pH.

Primary macronutrient levels in the soils are very high

- Primary macronutrients included in this test are potassium (K) and phosphorus (P)
- At the current cation exchange capacity, optimal K is around 175-215 ppm
- Optimal P is around 35-70 ppm
- **Potassium**
  - Average: 301 ppm
  - Min: 216 ppm – the minimum value here starts above the high end of the optimal range
  - Max: 345 ppm
- **Phosphorus**
  - Average: 293 ppm
  - Min: 216 ppm – this minimum value is more than 2x the high end of the optimal range
  - Max: 345 ppm
- **Recommendation:** limit application of additional P and K, no additional compost is needed. Remove plant biomass after each season and plant a cereal cover crop. This excess phosphorus can bind to iron and zinc leaving it less plant available.

Secondary macronutrients are also high, except for sulfur

- Secondary macronutrients included in this test are calcium (Ca), magnesium (Mg), and sulfur (S)
- Optimal Ca ranges from 2000-2500 ppm; Mg from 120-180 ppm; and S from 15-30 ppm
- **Calcium**
  - Average: 5126.8 ppm – this is over double of the high end of the optimal range
  - Min: 4232.5 ppm
  - Max: 6329.9 ppm
- **Magnesium**
  - Average: 363.1 ppm – this is double of the high end of the optimal range
  - Min: 320 ppm
  - Max: 423 ppm
- **Sulfur**
  - Average: 19.9 ppm
  - Min: 14.4 ppm
  - Max: 29.3 ppm
- **Recommendation:** Do not amend the soil with compost and plant nutrient heavy crops (such as brassicas) to help draw these levels down.

Micronutrients show mixed results

- Micronutrients included in the test are zinc (Zn) and Copper (Cu)
- Optimal Zn ranges from 8-10 ppm while optimal copper ranges from 1.0-5.8 ppm and Zinc levels should be double copper levels.
- **Zinc**
  - Average: 21.9 ppm
  - Min: 17.4 ppm
  - Max: 28.7 ppm
- **Copper**
  - Average: 2.1 ppm
  - Min: 1.8 ppm
  - Max: 2.6 ppm
- **Recommendation:** Consider adding copper sulfate to soils to try and bring levels up to 4-5 ppm

Cation exchange capacity shows the soil has a high capacity for nutrient retention and base saturation percentages are good

- **CEC**
  - Average: 18.8 meq/100g
  - Min: 18.1 meq/100g
  - Max: 19.5 meq/100g
- **Base Saturation of K – potassium should be the least dominant one in base saturation**
  - Average: 4.1%
  - Min: 2.5%
  - Max: 6.7%
- **Base Saturation of Mg – magnesium should be the second dominant one in base saturation**
  - Average: 16.1%
  - Min: 14.7%
  - Max: 18.2%
- **Base Saturation of Ca – calcium should be the dominant one in base saturation**
  - Average: 79.8%
  - Min: 77.1%
  - Max: 82.8%

Organic matter percentages are very high

- Optimal Ranges are between 3-5%
- Average: 10.6%
- Min: 7.4%
- Max: 13.5%
- **Recommendation:** The high organic matter can cause issues with root crops such as carrots, parsnips, and radishes however it can help with moisture retention for crops like tomatoes, cucumbers, and leafy greens

Conclusions

- Overall, garden levels of **nutrients are high**, **plots do not need compost amendments this year**.
- Only fertilizer suggestion would be copper sulfate
- **Ideal Crops:** Leafy greens, asparagus, brassicas (broccoli, kale, cabbage, etc.), and tomatoes
- **Non-Ideal Crops:** Root crops (carrots, potatoes, parsnips, etc.), and berries as well as other acid-loving crops
- Suggest cover cropping with cereal crops after growing season